

Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 6

Find the series' interval of convergence and, within this interval, the sum of the series as a function of x .

$$\sum_{n=0}^{\infty} \left(\frac{x^2 + 6}{9} \right)^n$$

First notice that this series is a geometric series.

Equate the terms of the geometric series $\sum_{n=0}^{\infty} ar^n$ to the given series $\sum_{n=0}^{\infty} \left(\frac{x^2 + 6}{9} \right)^n$ to find expressions for a and r in terms of x .

$$a = 1 \text{ and } r = \frac{(x^2 + 6)}{9}$$

The geometric series $\sum_{n=0}^{\infty} ar^n$ converges if $|r| < 1$ and diverges if $|r| \geq 1$.

So, the value of a does not affect the interval of convergence. To find the interval of convergence, use the convergence condition for the geometric series and substitute $\frac{(x^2 + 6)}{9}$ for r .

$$\begin{aligned} |r| &< 1 \\ \left| \frac{(x^2 + 6)}{9} \right| &< 1 \end{aligned}$$

Notice that the expression $\frac{(x^2 + 6)}{9}$ is nonnegative for all x , so the absolute value can be removed in order to simplify the solution.

$$\begin{aligned} \left| \frac{(x^2 + 6)}{9} \right| &< 1 \\ \frac{(x^2 + 6)}{9} &< 1 \end{aligned}$$

$$|a| = a \text{ if } a \geq 0.$$

$$x^2 + 6 < 9$$

Multiply both sides by 9.

$$x^2 < 3$$

Subtract 6 from both sides.

Solve the quadratic inequality for x by taking the square root of both sides of the inequality and rewriting the result as a compound inequality.

$$\begin{aligned} x^2 &< 3 \\ -\sqrt{3} &< x < \sqrt{3} \end{aligned}$$

Therefore the interval of convergence is $(-\sqrt{3}, \sqrt{3})$.

Recall that the series $\sum_{n=0}^{\infty} \left(\frac{x^2 + 6}{9} \right)^n$ was equated to the geometric series $\sum_{n=0}^{\infty} ar^n$ with $a = 1$ and $r = \frac{(x^2 + 6)}{9}$. Use the value of the infinite

geometric series to find an expression for $\sum_{n=0}^{\infty} \left(\frac{x^2 + 6}{9} \right)^n$.

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \quad |r| < 1$$

$$\sum_{n=0}^{\infty} \left(\frac{x^2 + 6}{9} \right)^n = \frac{1}{1 - \frac{x^2 + 6}{9}}, \quad -\sqrt{3} < x < \sqrt{3}$$

Substitute.

$$= \frac{9}{3 - x^2}, \quad -\sqrt{3} < x < \sqrt{3}$$

Simplify.